Hybrid Variable Elimination Ordering: v2 Results

TTIC 31180 — Probabilistic Graphical Models Final Project

What Changed in v2

Version 2 keeps the v1 baselines and adds three improvements: canonical `pgmpy` Bayesian-network benchmarks with native cardinalities, a redesigned fill-edge-local pressure term, and diagnostic cheap lookahead / bounded-lookahead runs. The original v1 hybrid penalized all high-degree neighbors; the v2 fill-pressure score penalizes only pressure created by actual fill-in edges, which is intended to reduce brittleness on hub-dominated graphs. The adaptive method chooses λ from graph statistics rather than from topology labels.

Setup

The v2 run used 71 graph instances and 872 successful method-instance runs. It loaded 5 canonical `pgmpy` Bayesian networks. Explicit pairwise VE completed for 478 of 872 rows under the factor-size cutoff. The cheap one-step method was run on 15 medium/small rows; bounded-lookahead oracle diagnostics produced 5 rows.

Benchmark	n	Edges	Max card.	Source
alarm	37	65	4	pgmpy_example_model
asia	8	10	2	pgmpy_example_model
child	20	30	6	pgmpy_example_model
hailfinder	56	99	11	pgmpy_example_model
insurance	27	70	5	pgmpy_example_model

Structural Results

Lower is better. Tables report topology-level medians.

Topology	Min-degree	Min-fill	Weighted	V1 degree	V2 fill	Adaptive
Low-treewidth	1.0	1.0	1.0	1.0	1.0	1.0
Random	22.0	20.0	21.0	21.0	21.0	21.0
Grid	11.5	11.5	11.5	11.0	12.0	11.5
Scale-free	10.0	10.0	10.0	10.0	12.0	10.0
Moralized BN	4.0	4.0	4.0	4.0	4.0	4.0

Topology	Min-degree	Min-fill	Weighted	V1 degree	V2 fill	Adaptive
Low-treewidth	1.40	1.40	1.40	1.40	1.40	1.40
Random	10.10	10.14	10.07	10.20	8.83	10.07
Grid	5.90	5.83	5.08	4.95	4.44	5.08
Scale-free	6.72	6.99	6.72	7.22	6.85	6.72
Moralized BN	2.33	2.33	2.33	2.33	2.33	2.33

Lambda Sensitivity

The fill-edge-local pressure term shifted the useful λ regime. In this run, larger λ values helped random and grid graphs more than the original v1 degree-pressure score, while low-treewidth and canonical BN benchmarks still preferred little or no extra pressure.

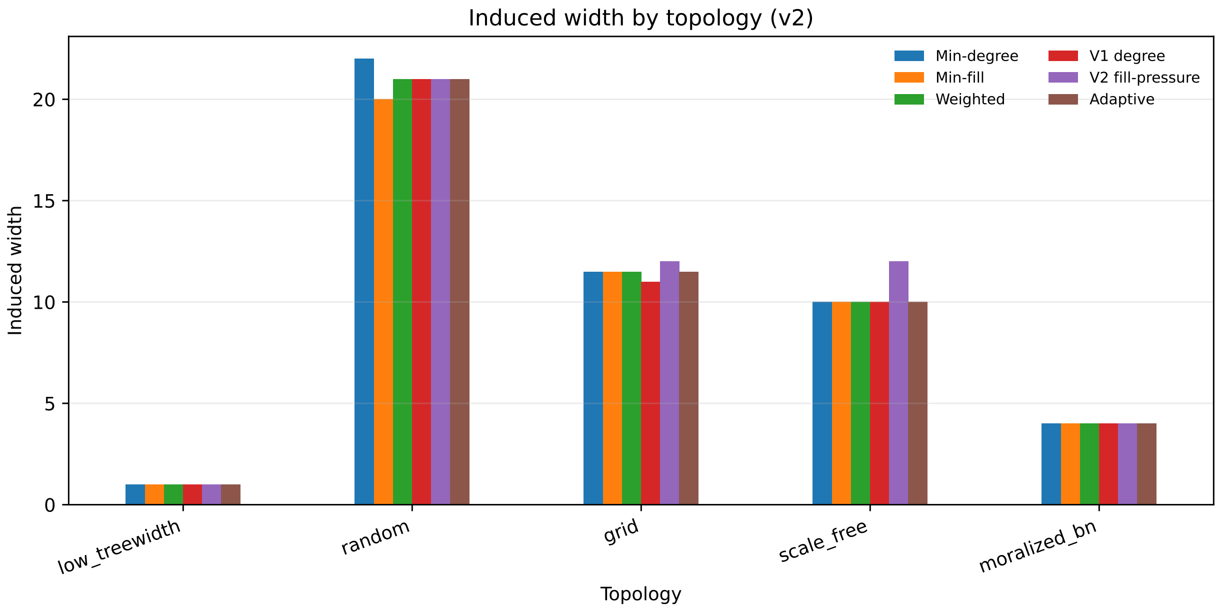
Topology	**Best λ**	**Median $\log_{10}$ peak size**
Low-treewidth	0.00	1.40
Random	1.00	8.83
Grid	1.00	4.44
Scale-free	0.05	6.62
Moralized BN	0.00	2.33

Failure Cases

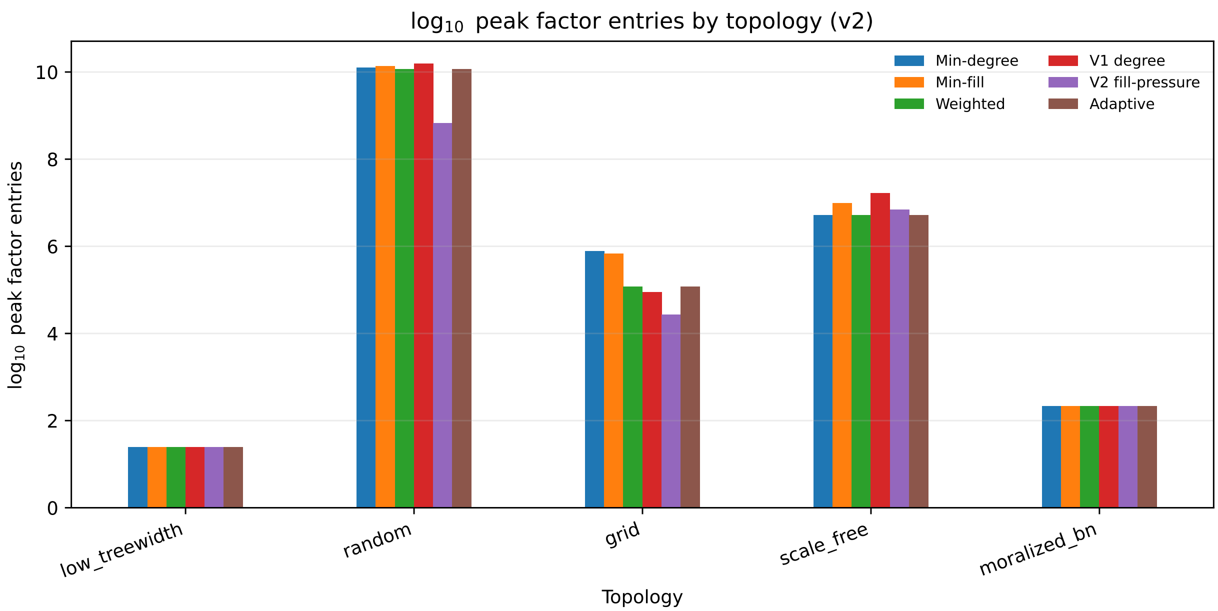
The table reports $\Delta = \log_{10}(\text{method peak}) - \log_{10}(\text{weighted min-fill peak})$ for the worst v1-degree failures. Negative values are improvements over weighted min-fill; positive values are worse.

Topology	**Graph**	**V1 Δ**	**V2 Δ**	**Adaptive Δ**
Random	er_n120_p0.10_s1	1.72	2.02	0.00
Random	er_n120_p0.06_s2	1.05	0.83	0.00
Scale-free	ba_n150_m2_s2	0.80	0.45	0.00
Moralized BN	pgmpy_insurance	0.78	0.60	0.78
Random	er_n80_p0.10_s1	0.75	0.13	0.00
Scale-free	ba_n100_m1_s0	0.70	0.00	0.00

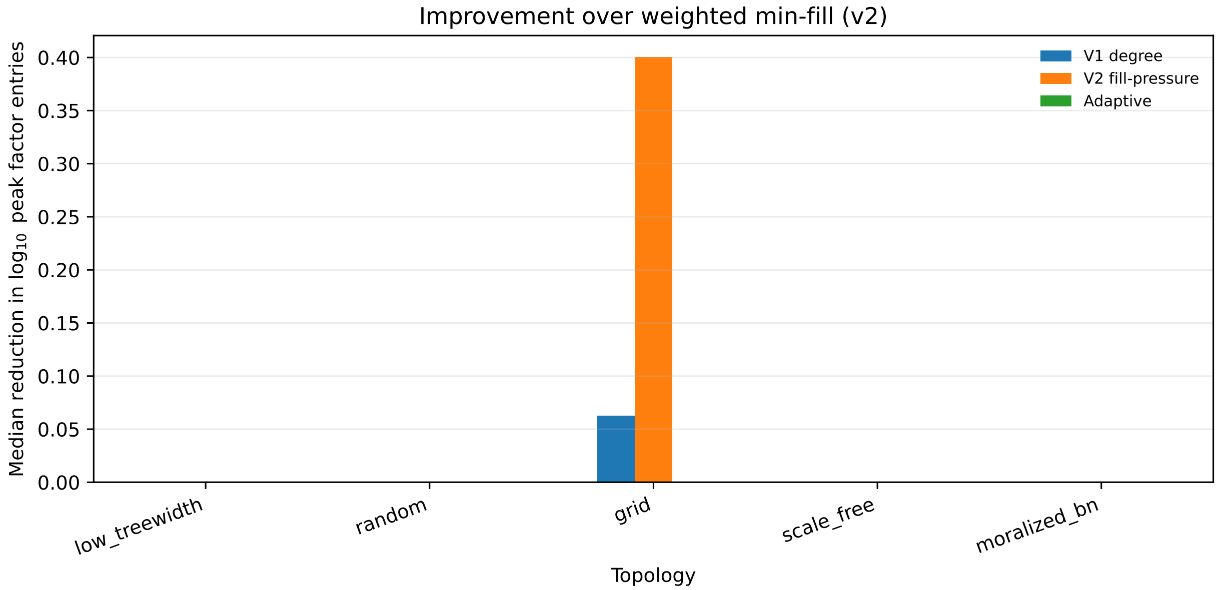
Figures

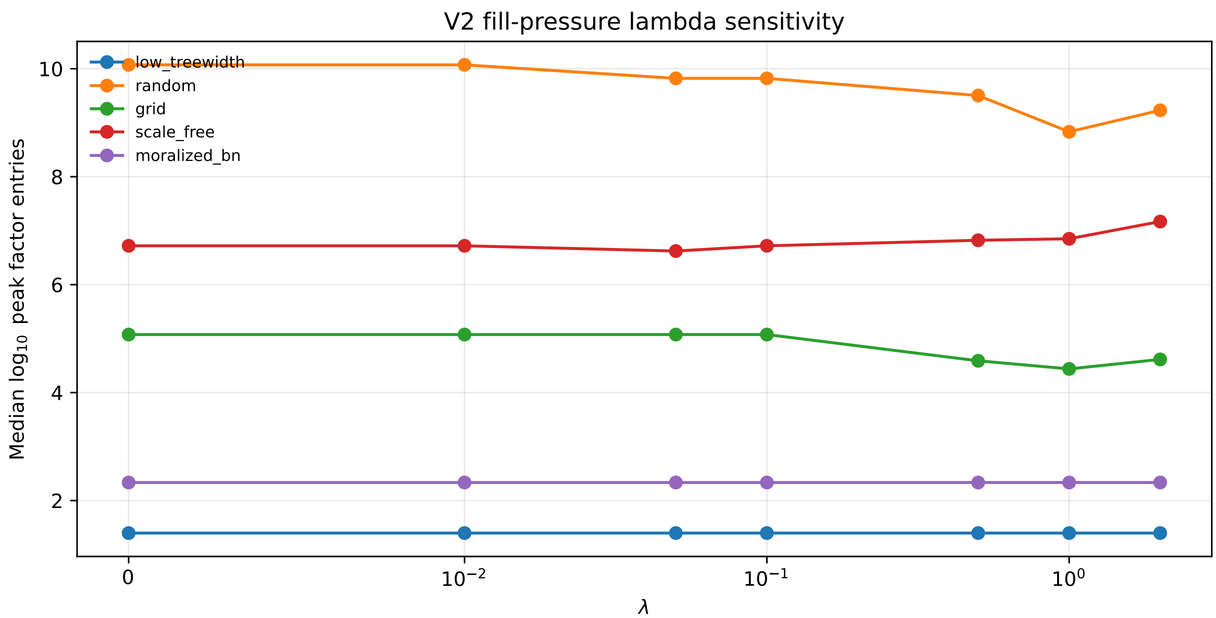


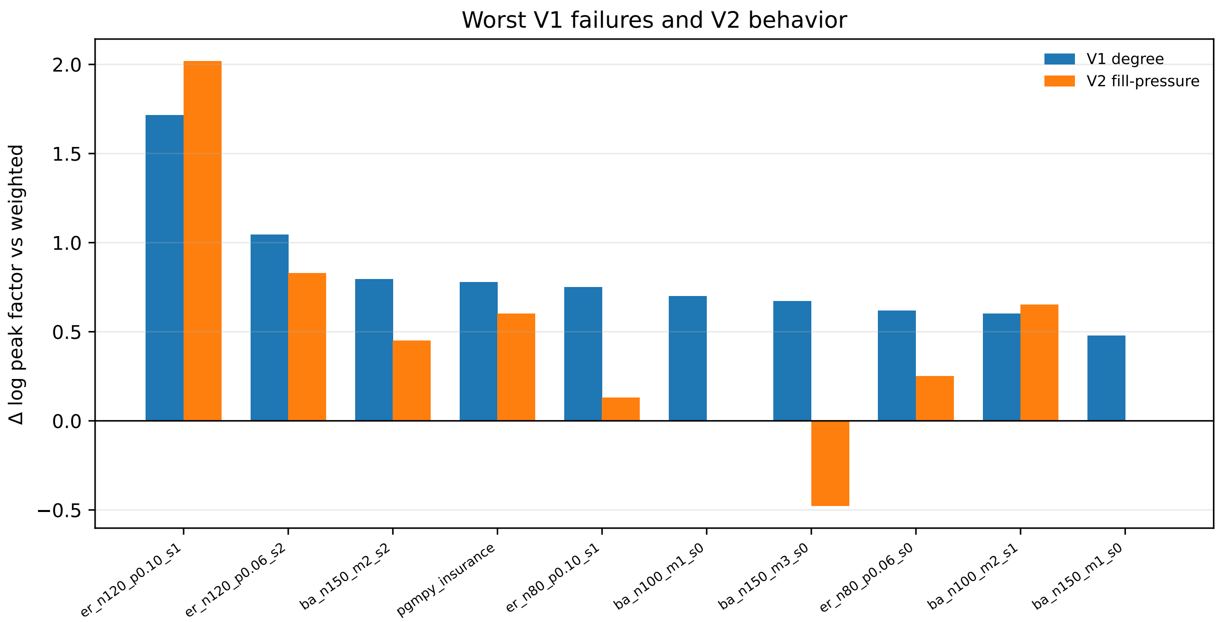
Induced width by topology (v2)
Min-degree
Min-fill
Weighted
V1 degree
V2 fill-pressure
Adaptive
Induced width
0
5
10
15
20
low_treewidth
random
grid
scale_free
moralized_bn
Topology



$\log_{10}$ peak factor entries by topology (v2)
Min-degree
Min-fill
Weighted
V1 degree
V2 fill-pressure
Adaptive
$\log_{10}$ peak factor entries
0
2
4
6
8
10
low_treewidth
random
grid
scale_free
moralized_bn
Topology


Improvement over weighted min-fill (v2)
Median reduction in $\log_{10}$ peak factor entries
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
V1 degree
V2 fill-pressure
Adaptive
low_treewidth
random
grid
scale_free
moralized_bn
Topology


V2 fill-pressure lambda sensitivity
Median $\log_{10}$ peak factor entries
2
4
6
8
10
low_treewidth
random
grid
scale_free
moralized_bn
0
10^{-2}
10^{-1}
10^{0}
λ



Findings and Implications

- The topology dependence remains the dominant story. No hybrid method uniformly dominates weighted min-fill.
- The redesigned fill-pressure term reduces one weakness of v1: it no longer broadly penalizes high-degree neighborhoods when the actual new fill-in is small.
- A fixed v2 $\lambda = 1$ gives stronger median peak-factor results on random and grid graphs than the conservative $\lambda = 0.1$ choice from v1, but it is still not a universal setting.
- Canonical BN benchmarks are less favorable to hybrid pressure: weighted min-fill is already strong, and extra pressure often adds little.
- Adaptive λ is useful as a concept, but the current hand-written rule is not yet reliable enough to replace a sweep. It avoids some v1 hub failures but can still misclassify random or BN-like structure.

Limitations

The v2 run improves benchmark realism, but it still uses structural pairwise VE instrumentation rather than full CPD-aware BN inference. The next-step heuristic is diagnostic-only on medium/small graphs because full-size one-step simulation was too expensive. The adaptive rule is intentionally simple and should be treated as a prototype rather than a tuned model-selection procedure.

Next Steps

- Replace the hand-written adaptive rule with a small validation-based selector that chooses λ from pilot instances by graph statistics.
- Add CPD-aware exact inference for canonical BNs, or use pgmpy/pyAgrum inference as a runtime validation layer.
- Expand oracle diagnostics on selected small graphs to compare cheap proxies against explicit lookahead more systematically.
- Analyze individual elimination traces for the random and insurance failure cases where adaptive fill-pressure is worse than weighted min-fill.
- Consider a guarded hybrid rule: use weighted min-fill by default, activate pressure only when pilot graph statistics predict a benefit.